

PATENT RESEARCH REPORT

1. ABSTRACT

This technical report synthesizes patent evidence for Hydrogenated Acrylonitrile NBR (HNBR), analyzing 12 primary patents published between 2009 and 2021. The research scope focuses on the catalytic hydrogenation of nitrile rubbers and the preparation of low molecular weight or functionalized HNBR variants. Key technical approaches include metathesis-mediated degradation using ruthenium-based catalysts (such as Grubbs, Hoveyda, and Arlt complexes) to control molecular weight, and heterogeneous catalytic systems (such as Raney Cobalt and supported Palladium) for hydrogenation and functionalization. Process trends reveal a strong emphasis on achieving low gel content (<5%), controlled Mooney viscosity, and introducing aminomethyl groups to enable alternative dianhydride-based crosslinking pathways. This report provides a comprehensive cross-patent comparison of reaction conditions, catalyst efficiencies, and resulting polymer properties.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

Patent Details - Patent Number: US7666950 - Patent Title: Process for preparing hydrogenated nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2010 - Material Type: Hydrogenated Nitrile Rubber (HNBR) - Relevance to target: Describes a preparation process for hydrogenated nitrile rubbers involving viscosity testing of diluted samples.

Preparation Method - Total Solids Dilution: 4 % (Diluted prior to viscosity testing)

Relevant Experimental Evidence - Examples listed in Table 2 were diluted to 4% total solids prior to viscosity testing.

Technical Relevance Provides methodology for preparing and testing the viscosity of hydrogenated nitrile rubber solutions.

Patent 2

Patent Details - Patent Number: EP1862477 - Patent Title: Process for preparing hydrogenated nitrile rubbers - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2009 - Material Type: Hydrogenated Nitrile Rubber (HNBR) - Relevance to target: Relates to a process for preparing hydrogenated nitrile rubbers.

Preparation Method

Relevant Experimental Evidence - No specific experimental parameters were disclosed in the extracted text.

Technical Relevance Contributes to the patent landscape of HNBR preparation processes.

Patent 3

Patent Details - Patent Number: US9079979 - Patent Title: Process for the preparation of hydrogenated nitrile rubber - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2015 - Material Type: Hydrogenated Nitrile Rubber (HNBR) - Relevance to target: Compares hydrogenation performance of Grubbs II, Grubbs III, and Wilkinson's catalysts.

Hydrogenation Method - Catalyst Type: Grubbs III / Grubbs II / Wilkinson's (Comparison of hydrogenation catalysts) - Catalyst Amount: 0.055 - 0.080 phr (HNBR 1 to 4) - Metal Content: 65.6 - 91.3 ppm (Catalyst metal content) - Reaction Time: 120 min (Grubbs III hydrogenates faster in first 120 min)

Relevant Experimental Evidence - Grubbs III catalyst at 0.057 phr (65.6 ppm metal) achieved 0.4% residual double bonds (RDB) after 120 min. - Wilkinson's catalyst at 0.06 phr achieved 3.1% RDB after 120 min. - Mooney viscosity of the resulting HNBR ranged from 20.2 to 69.3 depending on the catalyst and degradation.

Technical Relevance Demonstrates that Grubbs III catalyst provides faster hydrogenation kinetics than Grubbs II or Wilkinson's catalyst at equivalent metal contents.

Patent 4

Patent Details - Patent Number: EP2473280 - Patent Title: Process for the preparation of hydrogenated nitrile rubber - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2014 - Material Type: Hydrogenated Nitrile Rubber (HNBR) - Relevance to target: Equivalent to US9079979, detailing hydrogenation profiles and catalyst comparisons.

Hydrogenation Method - Catalyst Type: Grubbs III / Grubbs II / Wilkinson's (Comparison of hydrogenation catalysts) - Catalyst Amount: 0.057 - 0.080 phr (HNBR 1 to 4) - Metal Content: 65.6 - 91.3 ppm (Catalyst metal content)

Relevant Experimental Evidence - Grubbs III catalyst (HNBR 1) achieved 0.4% RDB in 120 min compared to 3.1% for Wilkinson's (HNBR 3). - Mooney viscosity of HNBR 1 was 38.2, while HNBR 3 was 69.3. - Grubbs III catalyst hydrogenates faster in the first 120 min than Grubbs II or Wilkinson's catalyst.

Technical Relevance Confirms the superior hydrogenation efficiency of Grubbs III metathesis catalysts in European jurisdictions.

Patent 5

Patent Details - Patent Number: US9150669 - Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2015 - Material Type: Low Molecular Weight HNBR - Relevance to target: Focuses on low molecular weight HNBR preparation using Hoveyda and Arlt catalysts.

Synthesis Method - Catalyst Type: Hoveyda / Arlt / Wilkinson's (Hydrogenation catalyst comparison) - Catalyst Amount: 0.041 - 0.060 phr (HNBR 1 to 3) - Reaction Time: 240 min (Total reaction duration)

Relevant Experimental Evidence - Hoveyda catalyst at 0.041 phr achieved 0.6% RDB in 120 min and 0.4% RDB in 240 min. - Arlt catalyst at 0.045 phr achieved 4.5% RDB in 120 min and 3.5% RDB in 240 min. - Hoveyda catalyst (HNBR 1) yielded a Mooney viscosity of 33.3 and PDI of 2.34 after 240 min.

Technical Relevance Shows that Hoveyda-type catalysts perform more efficiently in terms of hydrogenation and molecular weight reduction than Arlt or Wilkinson's catalysts.

Patent 6

Patent Details - Patent Number: EP2473281 - Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2014 - Material Type: Low Molecular Weight HNBR - Relevance to target: Details the dissolution and high-pressure hydrogenation process parameters for low molecular weight HNBR.

Preparation Method - Mooney Viscosity of Starting NBR: 24 - 70 MU (ML 1+4 at 100 °C) - Hydrogenation Temperature: 60 - 200 °C (Preferably 100 to 140 °C) - Dissolution Time: 12 hours (Dissolved in monochlorobenzene at room temperature) - Solution Concentration: 12 % (Transferred to high pressure reactor) - Reaction Pressure: 85 bar (Set pressure during hydrogenation)

Relevant Experimental Evidence - 518 g of nitrile rubber was dissolved in 4300 g of monochlorobenzene (12% solution) over 12 hours. - Hydrogenation was conducted at 138 °C and 85 bar pressure. - Hoveyda catalyst (0.041 phr) achieved 0.4% RDB after 240 min, yielding a polymer with Mw of 113,371 g/mol and Mooney viscosity of 33.3.

Technical Relevance Provides complete process parameters for dissolving NBR and executing high-pressure catalytic hydrogenation to obtain low molecular weight HNBR.

Patent 7

Patent Details - Patent Number: EP3827029 - Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2021 - Material Type: HNBR Compositions - Relevance to target: Focuses on

improving the aging properties of HNBR using supported catalysts and specific vulcanization conditions.

Preparation Method - Supported Catalyst Amount: 0.45 parts (Magnesium-silicate-supported metal catalyst comprising Pd) - Hydrogenation Temperature: 60 - 200 °C (Preferably 100 to 150 °C) - Vulcanization Temperature: 100 - 250 °C (Preferably 120 to 250 °C) - Vulcanization Time: 1 - 1440 min (Typically 1 min to 24 hours, preferably 2 min to 1 hour)

Relevant Experimental Evidence - Gel content of crude polymers increased by less than 10% during aging for 4 days at 140 °C. - Mooney viscosity increased by less than 40% during aging for 4 days at 140 °C. - Ruthenium compounds in the range of 10 ppm to 200 ppm showed only a small influence on polymer aging.

Technical Relevance Discloses methods to produce HNBR with improved thermal aging resistance, characterized by minimal gel content and Mooney viscosity increases.

Patent 8

Patent Details - Patent Number: EP2918612 - Patent Title: Hydrogenated nitrile rubber containing aminomethyl groups, method for producing ... - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2015 - Material Type: Aminomethyl-functionalized HNBR - Relevance to target: Introduces aminomethyl groups into HNBR to enable alternative crosslinking chemistry.

Synthesis Method - Hydrogenation Pressure: 50 - 250 bar (Preferably 50 to 120 bar) - Hydrogenation Temperature: 60 - 190 °C (Preferably 100 to 180 °C) - Aminomethyl Group Content: 0.01 - 20 wt% (Preferably 1 to 5 wt%) - Gel Content: < 10 % (Preferably less than 5%)

Relevant Experimental Evidence - Raney Cobalt catalyst (Grace 2724) was pretreated by washing with methanol and THF to remove water. - Hydrogenation of Therban 3406 with ammonia (20 g) at 130 °C and 120 bar yielded 2.4 wt% aminomethyl groups and a gel content of 2.4%. - Crosslinking with pyromellitic dianhydride (26.3 phr) and DBU (2.21 phr) at 170 °C yielded vulcanizates with high tensile strength (27.1 MPa) and microhardness (83 IRHD).

Technical Relevance Enables the synthesis of aminomethyl-functionalized HNBR with low gel content, which can be crosslinked using dianhydrides instead of peroxides or sulfur.

Patent 9

Patent Details - Patent Number: EP2918608 - Patent Title: Hydrogenated nitrile rubber containing amino methyl group and method for ... - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2015 - Material Type: Aminomethyl-functionalized HNBR - Relevance to target: Equivalent or closely related to EP2918612, detailing aminomethyl functionalization and crosslinking.

Synthesis Method - Hydrogenation Pressure: 50 - 250 bar (Preferably 50 to 120 bar) - Hydrogenation Temperature: 60 - 190 °C (Preferably 100 to 180 °C) - Aminomethyl Group Content: 0.01 - 20 wt% (Preferably 1 to 5 wt%)

Relevant Experimental Evidence - Therban 3406 was hydrogenated using pretreated Raney Cobalt and ammonia at 130 °C and 120 bar, achieving 11.6% nitrile conversion (2.2 wt% aminomethyl content). - Addition of n-dodecyl sulfide (0.1 wt% relative to catalyst) improved conversion and controlled gel formation. - Crosslinking with maleic anhydride at 180 °C for 2 hours achieved a gelation degree of 57.1% to 68.4%.

Technical Relevance Confirms the reproducibility of aminomethyl functionalization of HNBR and demonstrates crosslinking with cyclic anhydrides like maleic anhydride.

Patent 10

Patent Details - Patent Number: US7951875 - Patent Title: Low molecular weight hydrogenated nitrile rubber - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2011 - Material Type: Low Molecular Weight HNBR - Relevance to target: Describes simultaneous metathesis degradation and hydrogenation of NBR to produce low molecular weight HNBR.

Synthesis Method - Polymer Concentration: 6 - 15 wt% (Dissolved in monochlorobenzene (MCB)) - Co-olefin Pressure: 500 psi (Ethylene (C₂H₄) pressure) - Reaction Temperature: 80 °C (Metathesis reaction temperature) - Catalyst Load: 0.05 - 0.25 phr (Metathesis catalyst loading)

Relevant Experimental Evidence - NBR rubber (200 g) was dissolved in 1133 g of MCB to form a 15 wt% solid solution. - Metathesis degradation in the presence of ethylene (500 psi) at 80 °C reduced Mn from 85 kg/mol to 27 kg/mol and Mw from 296 kg/mol to 54 kg/mol. - 1-hexene (150 g) was also successfully used as a co-olefin for degradation.

Technical Relevance Establishes a process for preparing low molecular weight HNBR via olefin metathesis degradation using co-olefins like ethylene or 1-hexene.

Patent 11

Patent Details - Patent Number: US7919563 - Patent Title: Low molecular weight hydrogenated nitrile rubber - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2011 - Material Type: Low Molecular Weight HNBR - Relevance to target: Equivalent to US7951875, detailing metathesis degradation parameters.

Synthesis Method - Polymer Concentration: 6 - 15 wt% (Dissolved in monochlorobenzene (MCB)) - Co-olefin Pressure: 500 psi (Ethylene (C₂H₄) pressure) - Reaction Temperature: 80 °C (Metathesis reaction temperature)

Relevant Experimental Evidence - NBR rubber (75 g) was dissolved in 1175 g of MCB to form a 6 wt% solid solution. - Metathesis degradation with 1-hexene (150 g) at 80 °C successfully reduced molecular weight. - Resulting polymers exhibited narrow polydispersity index (PDI) and significantly reduced Mooney viscosity.

Technical Relevance Confirms the viability of using alternative co-olefins (e.g., 1-hexene) and lower solution concentrations (6 wt%) for controlled NBR degradation.

Patent 12

Patent Details - Patent Number: EP2700692 - Patent Title: Hydrogenated nitrile rubber composition - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2019 - Material Type: HNBR Composition - Relevance to target: Focuses on compounding HNBR with specific fillers and crosslinking agents for LLC resistance.

Compounding Method - Bound Acrylonitrile Content: 31 - 50 % (HNBR polymer specification) - Iodine Value: ≤ 30 mg/100mg (HNBR polymer specification) - MT Carbon Black: 30 - 300 parts by weight (Based on 100 parts HNBR) - Cocrosslinking Agent: 1 - 30 parts by weight (Based on 100 parts HNBR)

Relevant Experimental Evidence - Compositions using 150 parts of MT carbon black and 6 parts of cocrosslinking agent showed excellent LLC (long-life coolant) resistance. - Silica (30 to 300 parts) and silane coupling agents (0 to 5 parts) can be used to achieve similar performance. - Roll processability was rated as excellent or good for the optimized filler and crosslinker ratios.

Technical Relevance Provides compounding formulations for HNBR to achieve superior resistance to long-life coolants (LLC) and good roll processability.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

Comparison Table: | Patent | Hydrogenation Catalyst | Key Finding | |---|---|---| | US7666950 | Not disclosed | | | EP1862477 | Not disclosed | | | US9079979 | Not disclosed | | | EP2473280 | Not disclosed | | | US9150669 | Not disclosed | | | EP2473281 | Not disclosed | | | EP3827029 | Not disclosed | | | EP2918612 | Not disclosed | | | EP2918608 | Not disclosed | | | US7951875 | Not disclosed | | | US7919563 | Not disclosed | | | EP2700692 | Not disclosed | |

Trend Analysis: - Catalyst Selection: Metathesis-based ruthenium catalysts (Grubbs II/III, Hoveyda, Arlt) are preferred for simultaneous degradation and hydrogenation to produce low molecular weight HNBR, whereas Raney Cobalt is utilized for functionalization (aminomethylation) and supported Pd is used for standard hydrogenation. - Process Conditions: Hydrogenation temperatures typically range from 80 °C to 160 °C, with pressures varying widely from 500 psi (approx. 34 bar) for metathesis degradation up to 120 bar for Raney Cobalt-catalyzed aminomethylation. - Molecular Weight Control: Metathesis degradation processes successfully reduce the weight-average molecular weight (Mw) from ~296 kg/mol down to ~54 kg/mol, significantly lowering Mooney viscosity to improve processability. - Functionalization and Crosslinking: Aminomethyl-functionalized HNBR (1-5 wt% amine content) enables alternative crosslinking with dianhydrides (e.g., pyromellitic dianhydride) at 170-180 °C, achieving high tensile strength (up to 27.1 MPa) without peroxide or sulfur.

4. CONCLUSION

The patent landscape for Hydrogenated Acrylonitrile NBR (HNBR) reveals a strong focus on process optimization, molecular weight reduction, and functionalization. Metathesis degradation using ruthenium-based catalysts (Grubbs, Hoveyda) is a dominant trend for producing low molecular weight HNBR with improved processability. Additionally, aminomethyl functionalization using Raney Cobalt catalysts introduces novel crosslinking pathways via dianhydrides, offering peroxide-free vulcanization options. Gaps in the disclosed patent data include detailed long-term dynamic mechanical properties and specific emulsifier/stabilizer formulations used during the initial polymerization stage.

5. REFERENCES

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